CWC-Scale Generator

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Ansys

Epoxy vs Super Glue

- 919 Super Glue
- ET5441 Epoxy

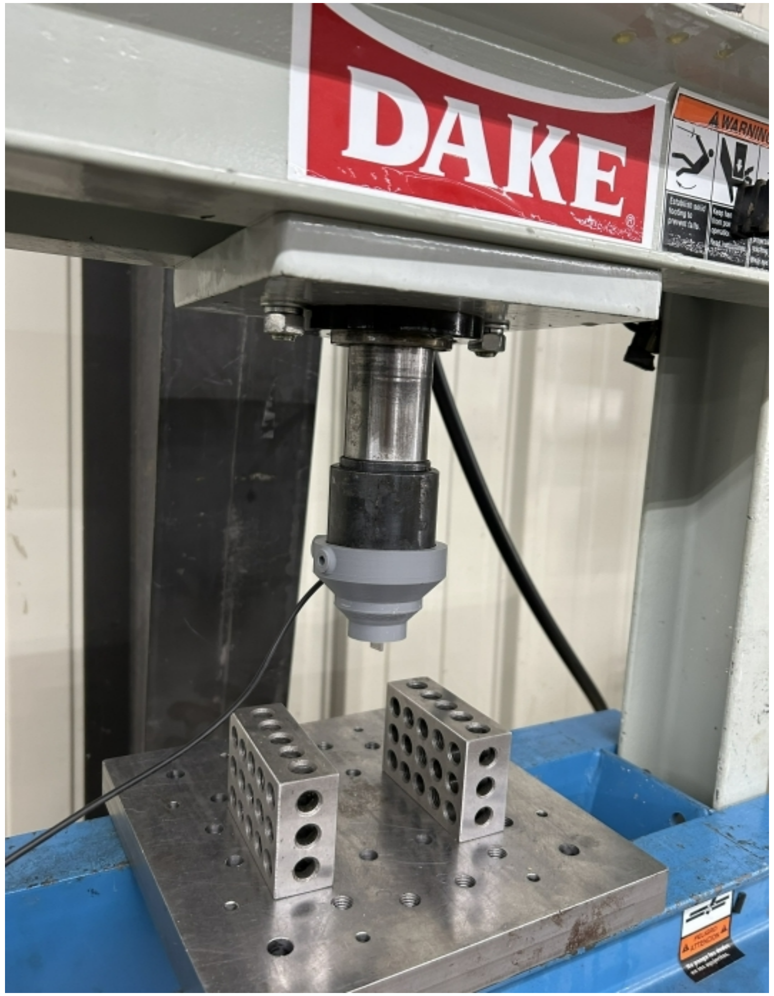


Figure 1: Pneumatic Press

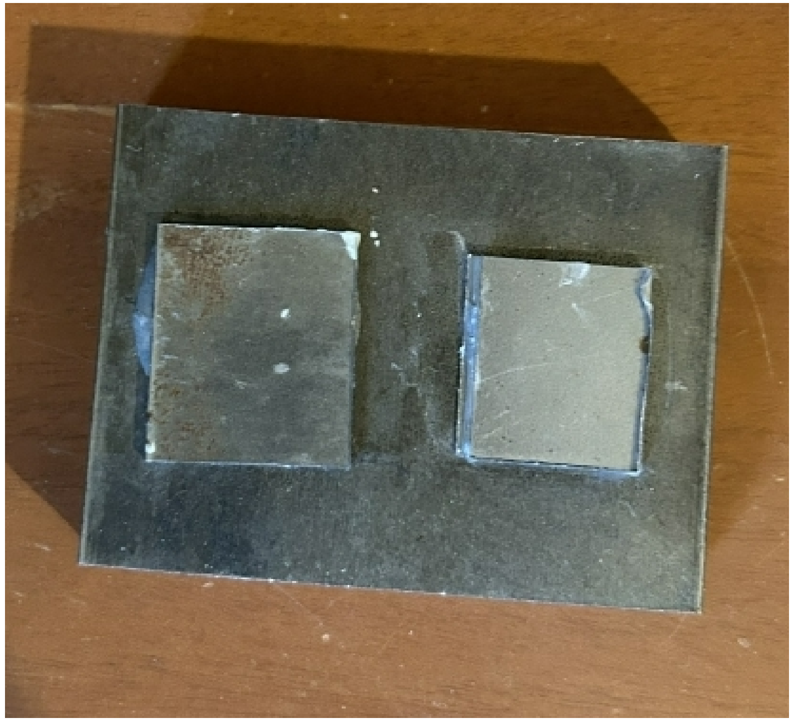


Figure 2: Super and Epoxy adhering steel

Epoxy vs Super Glue

Table 1: Epoxy Results

Trial	Force Applied Until Failure (lbs)
1	72.2
2	81.3
3	91.1
4	81.1
5	55.4

Table 2: Super Glue Results

Trial	Force Applied Until Failure (lbs)
1	126.1
2	133.4
3	167.3
4	187.2
5	143.3

Epoxy vs Super Glue

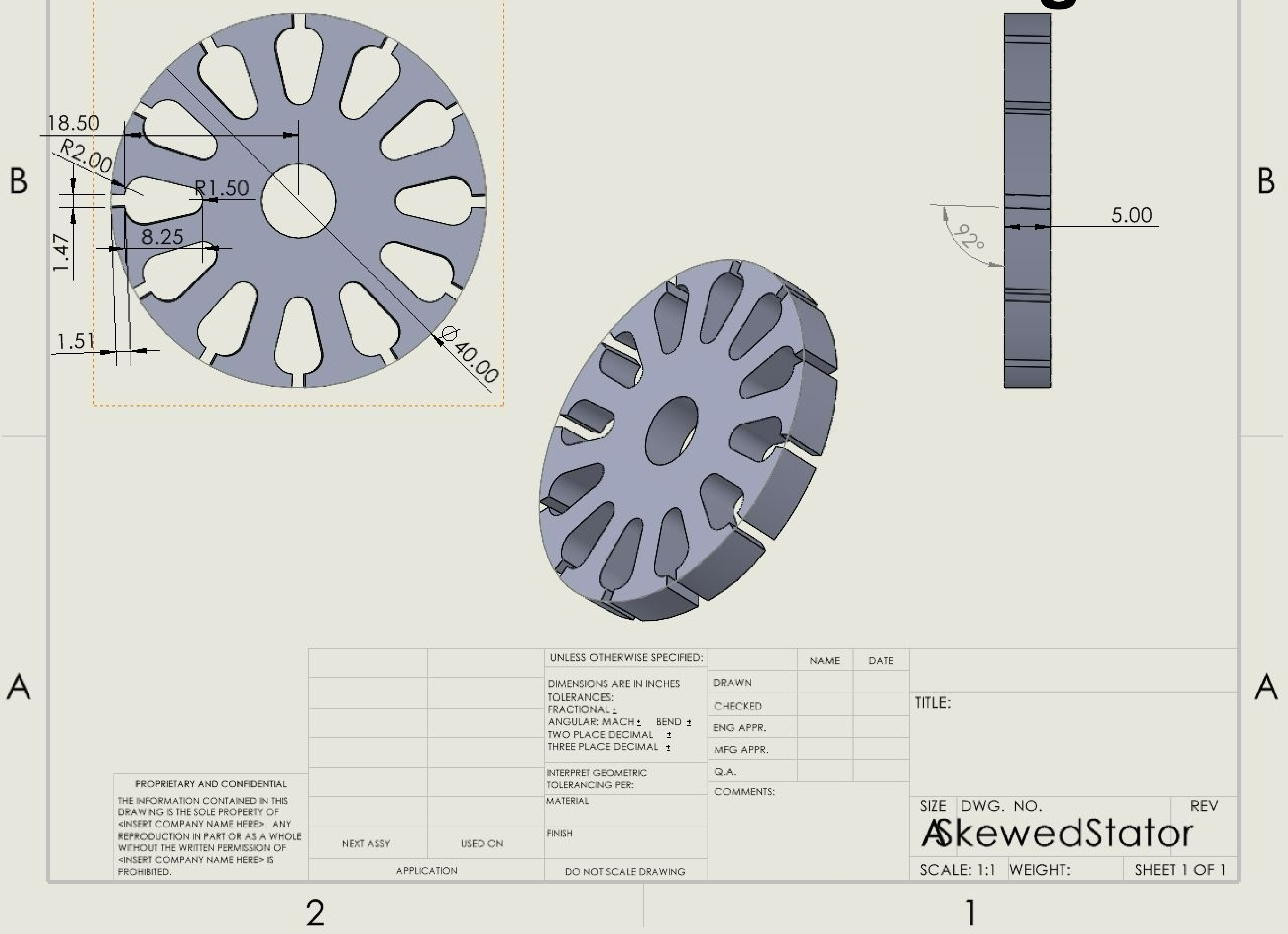
Epoxy

- Messy
- Difficult to apply
- Max force of 91.1 pounds
- Single use tubes
- Can withstand up to 356 degrees Fahrenheit

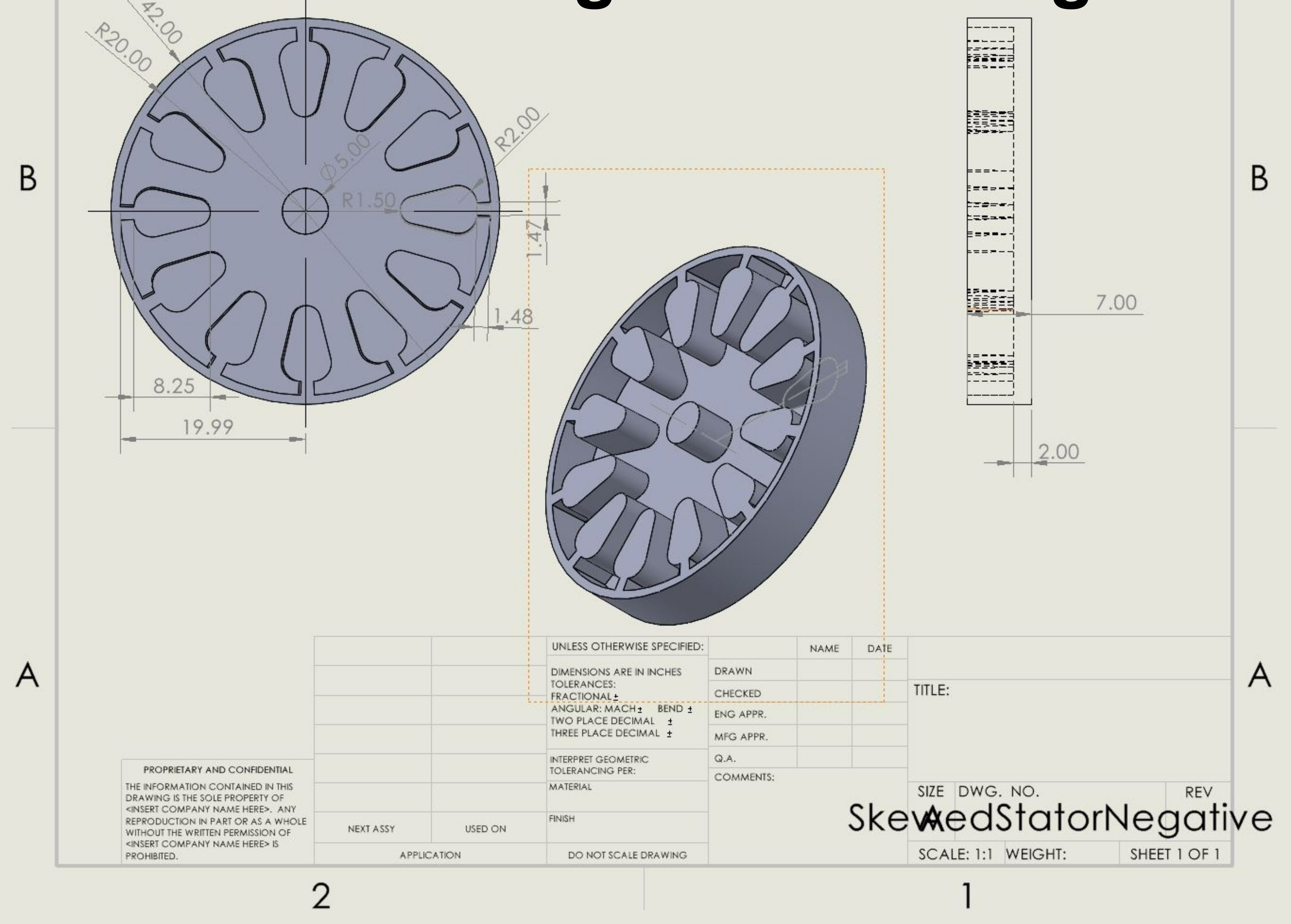
Super Glue

- Relatively Clean
- Easy to apply
- Requires a very little amount
- Max force of 187.2
- Can withstand up to 482 degrees Fahrenheit

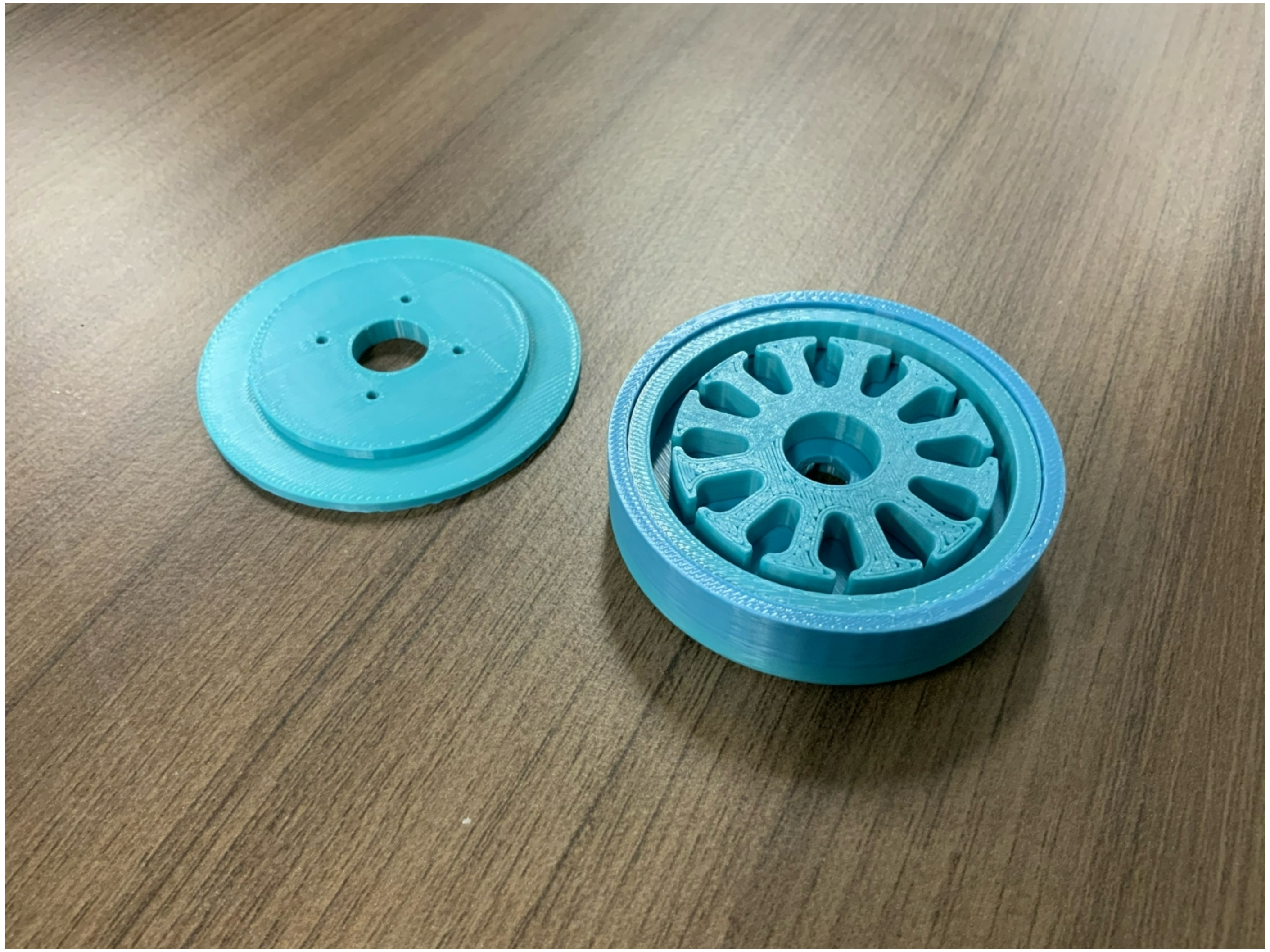
Skewed Stator Drawing



Stator Negative Drawing

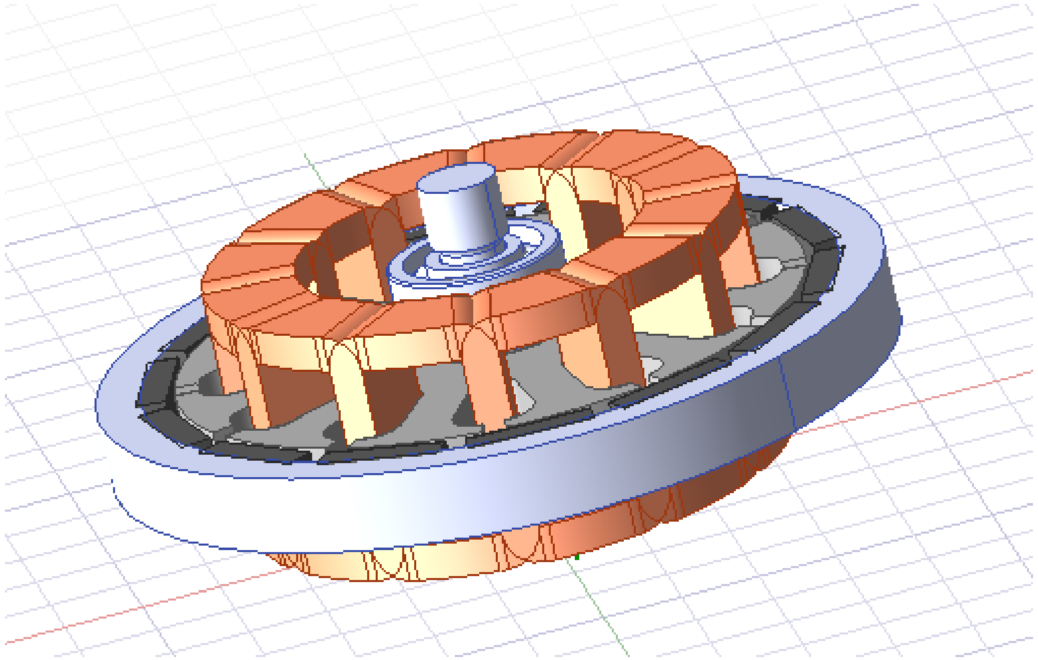


Physical Prototype

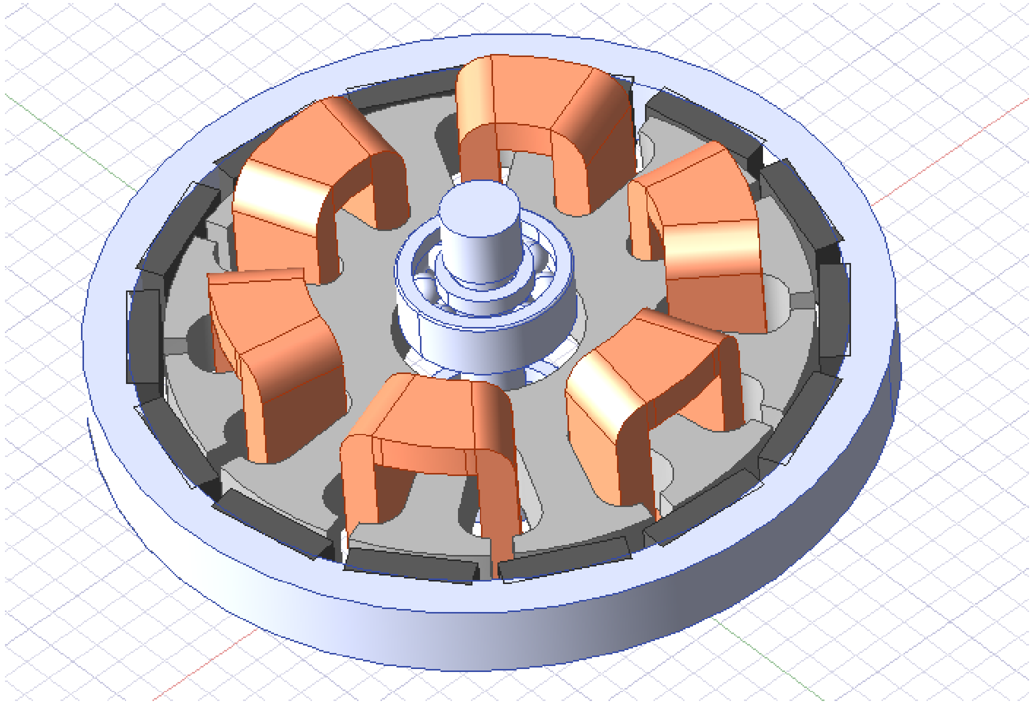


Ansys Simulation

Final CAD Model

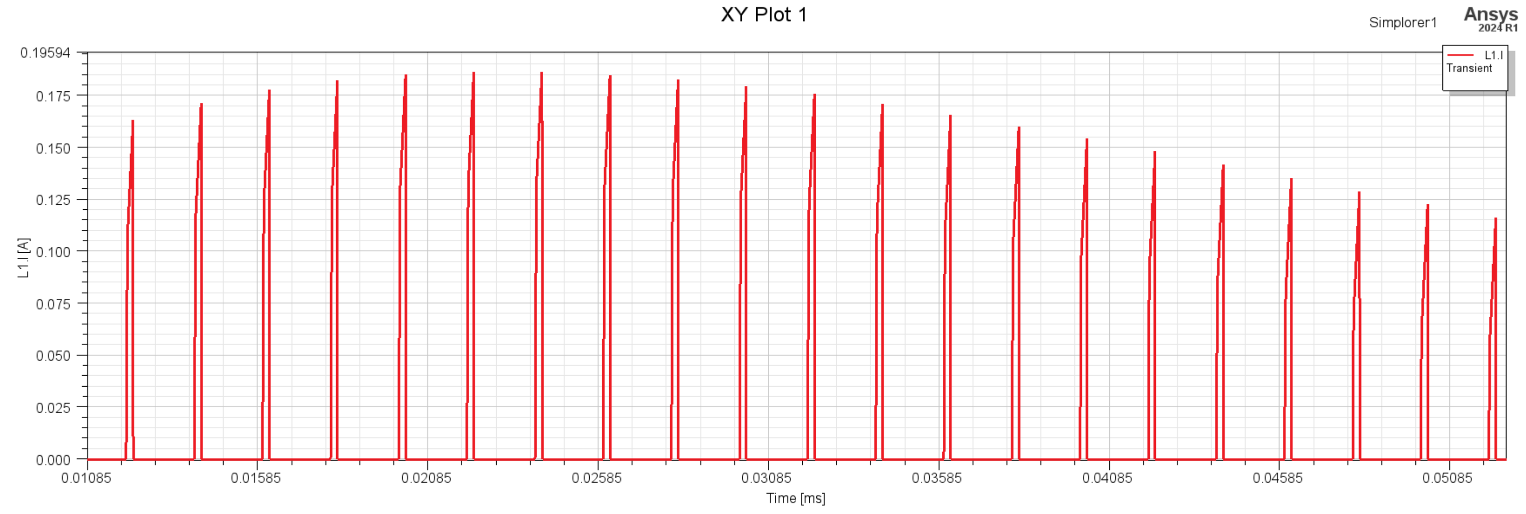


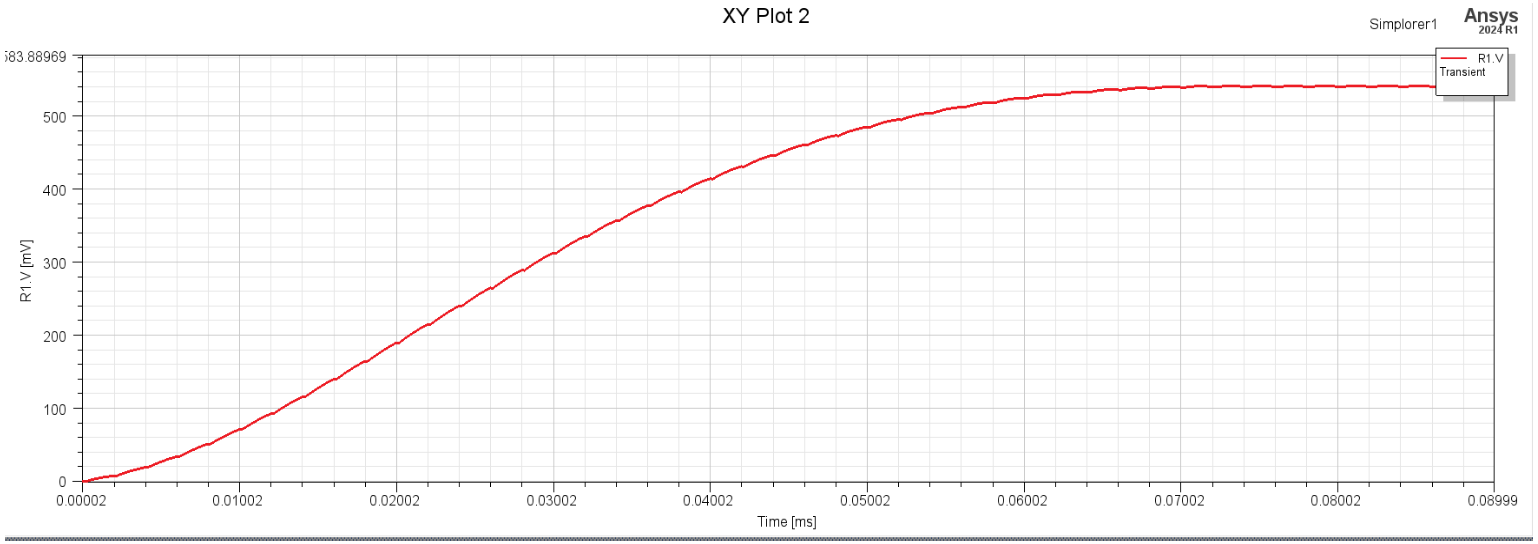
Modified for Simulations



Ansys Simulation Results

Current Plot





Voltage within the resistor

Ansys Simulation Errors

Error Messages received when trying to run Simulation

Message Manager

- *Global - Messages
- Project1 (1) (C:/Users/kpr77/Downloads/)
  - Maxwell3DDesign1 (Transient)
    - No part to be cleaned up. (03:32:54 PM Apr 28, 2025)
    - Verify conduction path: Conduction path validate success. (03:39:04 PM Apr 28, 2025)
    - Currently no mesh operations or data link have been created for the setup(s): Transient. The initial mesh will be used. The initial mesh is usually not fine enough to achieve an accurate solution. (03:39:04 PM Apr 28, 2025)
    - The number of computational domain doesn't equal to one. %1 (03:40:14 PM Apr 28, 2025)
    - Simulation completed with execution error on server: Local Machine. (03:40:15 PM Apr 28, 2025)
  - Simplorer1
  - Nothing to solve for Transient. Enable or clean up solution to force a new simulation. (03:39:02 PM Apr 28, 2025)

Thank you

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Any Questions?

References

[1] Climate Central, "A Decade of Growth in Solar and Wind Power: Trends Across the U.S. | Climate Central," *www.climatecentral.org*, Apr. 03, 2024. https://www.climatecentral.org/report/solar-and-wind-power-2024